Homework 2

CS/ECE 374 Introduction to Algorithms and Models of Computation

University of Illinois Urbana Champaign

Spring 2025

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Maximum Points: 100
Due Date: Monday, Feb. 10, 2025, 11:59 PM

Instructions

1. Answer **ALL** questions.
2. Show all the work.
3. Legibly scribe your solutions and upload them as individual problems on Gradescope.

Suggestions

While writing your answers and/or proofs, think about the following.

1. What is it that you are given? What is it that you are assuming? What is it that you are trying to prove?
2. How exactly does a step follow from the previous step? Assume nothing is apparent, even algebraic simplifications. Your reasoning should be clear and convincing.
3. If you are using a result from the class then state it clearly before using it. You should identify where exactly you need that result, cast your problem into the setup of the result, and then use it.
4. If you are using a result from outside the class then you should prove it before using it

Notations

- For a string x, $n_0(x)$ denotes the number of zeros in it and $n_1(x)$ denotes the number of ones in it.
- For a string x, $s(x)$ denotes the first element of x and $e(x)$ denotes the last element of x.

Definitions

Definition 1. *A binary string x is said to be divisible by a natural number n, if the decimal number associated with x is divisible by n. We denote x divisible by n as $x \bmod n = 0$.*

Questions

1. (a) Consider the following language. [15]

$$L = \{x : x \in \{0,1\}^* \text{ and } x \bmod 3 = 0\} \cup \{\epsilon\}.$$

Show that L is regular.

Solution. The automaton that recognizes L is given in Figure 1.

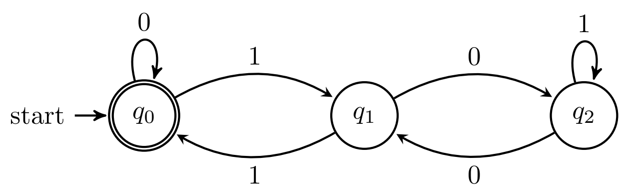


Figure 1: DFA recognizing L.

(b) Prove that the class of regular languages is closed under the complement operation. [5]

Solution. Suppose language B over alphabet Σ has a DFA $M = (Q, \Sigma, \delta, q_1, F)$. Then, a DFA for the complementary language $\overline{B}$ is $M' = (Q, \Sigma, \delta, q_1, Q - F)$.

The reason why M' recognizes $\overline{B}$ is as follows. First note that M and M' have the same transition function δ. Thus, since M is deterministic, M' is also deterministic.

Now consider any string $w \in \Sigma^*$. Running M on input string w will result in M ending in some state $r \in Q$. Since M is deterministic, there is only one possible state that M can end in on input w. If we run M' on the same input w, then M' will end in the same state r since M and M' have the same transition function. Also, since M is deterministic, there is only one possible ending state that M can be in on input w.

Now suppose that $w \in B$. Then M will accept w, which means that the ending state $r \in F$, i.e., r is an accept state of M. But then $r \in Q - F$, so M' does not accept w since M' has $Q - F$ as its set of accept states. Similarly, suppose that $w \notin B$. Then M will not accept w, which means that the ending state $r \notin F$. But then $r \in Q - F$, so M' accepts w. Therefore, M' accepts string w if and only if M does not accept string w, so M' recognizes language $\overline{B}$.

Hence, the class of regular languages is closed under complement.

2. Show that each of the following languages defined over $\{0,1\}$ is regular. [20]

(a) $L_1 = \{x : n_0(x) \bmod 2 = 0\}$.

(b) $L_2 = \{x : n_0(x) \bmod 2 = 1\}$.

(c) $L_3 = \{x : n_1(x) \bmod 2 = 0\}$.

(d) $L_4 = \{x : n_1(x) \bmod 2 = 1\}$.

Solution. The automata that recognize L_1, L_2, L_3, and L_4 are given in Figures 5, 2b, 2c, and 2d, respectively.

3. For each of the following languages defined over $\{0,1\}$, construct a deterministic finite automaton that recognizes it. Hence or otherwise show that each language is regular. [20]

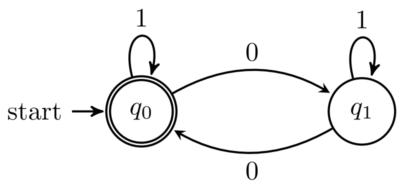
(a) $L_5 = \{x : n_0(x) \bmod 2 = 0 \text{ and } n_1(x) \bmod 2 = 0\}$.

(b) $L_6 = \{x : n_0(x) \bmod 2 = 1 \text{ and } n_1(x) \bmod 2 = 0\}$.

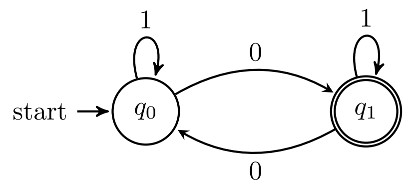
(c) $L_7 = \{x : n_0(x) \bmod 2 = 0 \text{ and } n_1(x) \bmod 2 = 1\}$.

(d) $L_8 = \{x : n_0(x) \bmod 2 = 1 \text{ and } n_1(x) \bmod 2 = 1\}$.

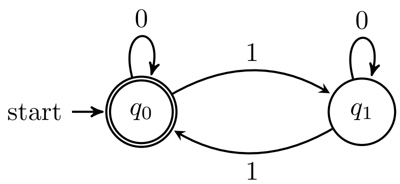
Solution. The automata that recognize L_5, L_6, L_7, and L_8 are given in Figures 3a, 3b, 3c, and 3d, respectively.



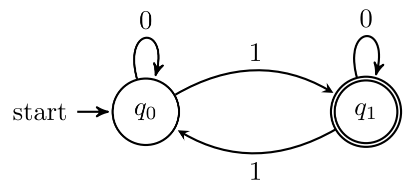
(a) DFA recognizing L_1.



(b) DFA recognizing L_2.



(c) DFA recognizing L_3.



(d) DFA recognizing L_4.

Figure 2: Solution to Question 4.

4. For each of the following languages defined over $\{0, 1\}$, construct a deterministic finite automaton that recognizes it. Hence or otherwise show that each language is regular. [20]

   (a) $L_9 = \{x : s(x) = 0 \text{ and } e(x) = 1\}$.

   (b) $L_{10} = \{x : s(x) \neq 0 \text{ or } e(x) \neq 1\}$.

   Solution. The automata that recognize L_9 and L_{10} are given in Figures 4a and 4b, respectively.

5. (a) Convert the NFA given in Figure 5 to an equivalent DFA. [15]

   Solution. For the given NFA, the corresponding finite automaton is formalized as follows.

   i. $Q = \{q_0, q_1, q_2\}$.
   ii. $\Sigma = \{0, 1\}$.
   iii. $\delta : Q \times \Sigma \to Q$ is the transition function given as follows.

δ	0	1
q_0	$\{q_0\}$	$\{q_0, q_1\}$
q_1	$\{q_2\}$	$\{q_2\}$
q_2	ϕ	ϕ

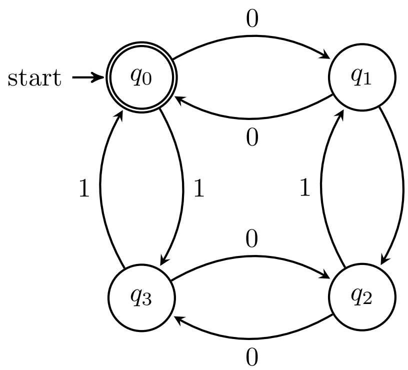
   iv. $q_0 = q_1$.
   v. $F = \{q_2\}$.

   The formal representation of the above NFA upon conversion to a DFA is as follows.

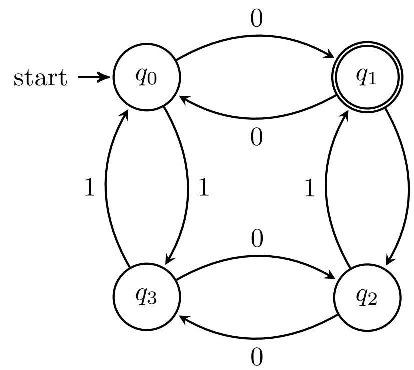
   i. $Q = \{\{q_0\}, \{q_1\}, \{q_2\}, \{q_0, q_1\}, \{q_0, q_2\}, \{q_1, q_2\}, \{q_0, q_1, q_2\}\}$.
   ii. $\Sigma = \{0, 1\}$.
   iii. $\delta : Q \times \Sigma \to Q$ is the transition function given as follows.

δ	0	1
$\{q_0\}$	$\{q_0\}$	$\{q_0, q_1\}$
$\{q_0, q_1\}$	$\{q_0, q_2\}$	$\{q_0, q_1, q_2\}$
$\{q_0, q_2\}$	$\{q_0\}$	$\{q_0, q_1\}$
$\{q_0, q_1, q_2\}$	$\{q_0, q_2\}$	$\{q_0, q_1, q_2\}$

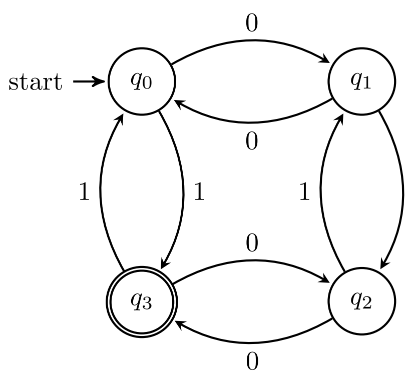
   iv. $q_0 = \{q_0\}$.
   v. $F = \{\{q_2\}, \{q_0, q_2\}, \{q_1, q_2\}, \{q_0, q_1, q_2\}\}$.



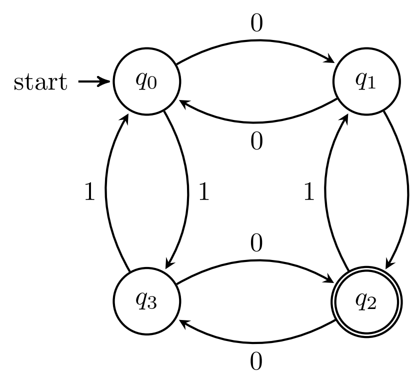
(a) DFA recognizing L_5.



(b) DFA recognizing L_6.



(c) DFA recognizing L_7.



(d) DFA recognizing L_8.

Figure 3: Solution to Question 5.

Refer to Figure 6 for the drawing of this equivalent DFA.

(b) Prove that the class of regular languages is closed under the star operation. [5]

Solution. Let $N_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognize A_1.

Construct $N = (Q, \Sigma, \delta, q_0, F)$ to recognize A_1^* as follows:

i. $Q = \{q_0\} \cup Q_1$. The states of N are the states of N_1 plus a new start state.

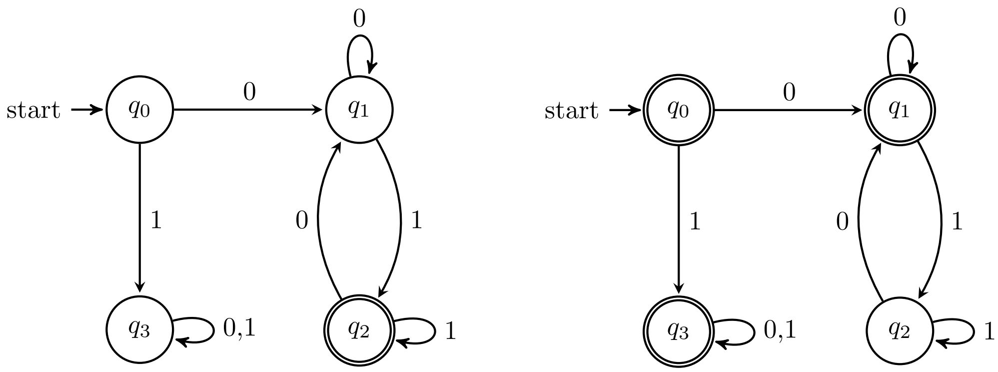
ii. The state q_0 is the new start state.

iii. $F = \{q_0\} \cup F_1$. The accept states are the old accept states plus the new start state.

iv. Define δ so that for any $q \in Q$ and any $a \in \Sigma_\epsilon$,

$$\delta(q,a) = \begin{cases} \delta_1(q,a) & \text{if } q \in Q_1 \text{ and } q \notin F_1 \\ \delta_1(q,a) \cup \{q_1\} & \text{if } q \in F_1 \text{ and } a = \epsilon \\ \{q_1\} & \text{if } q = q_0 \text{ and } a = \epsilon \\ \emptyset & \text{otherwise} \end{cases}$$

- The new start state q_0 allows the automaton to start processing any number of strings recognized by N_1.
- The new accept states include the old accept states plus q_0 to allow acceptance of the empty string.
- The transition function δ includes:
  - Transitions within N_1 when the current state is not an accept state.
  - Transitions that loop back to the start state q_1 when in an accept state with ϵ transitions, allowing concatenation.
  - Initial ϵ transition from the new start state q_0 to q_1.



(a) DFA recognizing L_9. (b) DFA recognizing L_{10}.

Figure 4: Solution to Question 6.

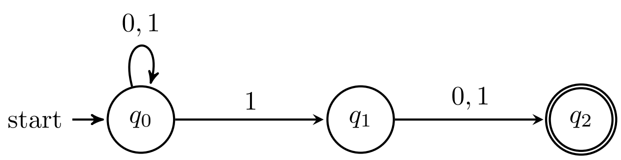


Figure 5

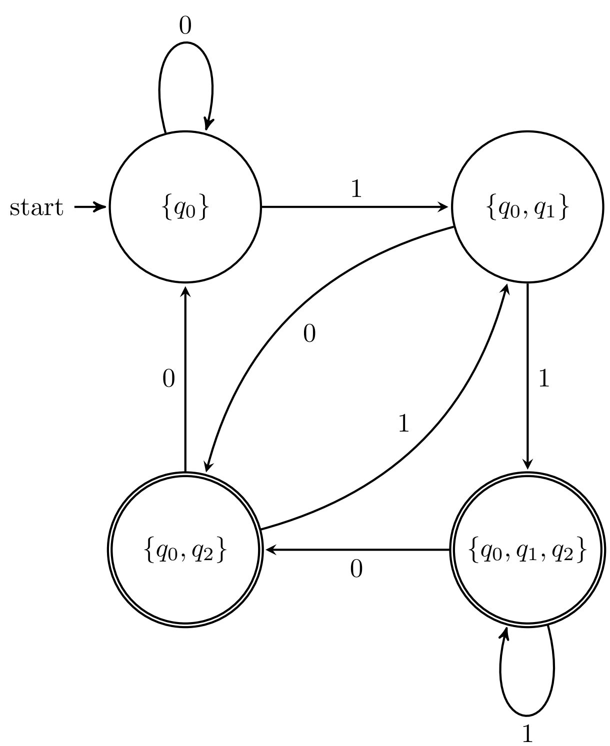


Figure 6: Automaton M.